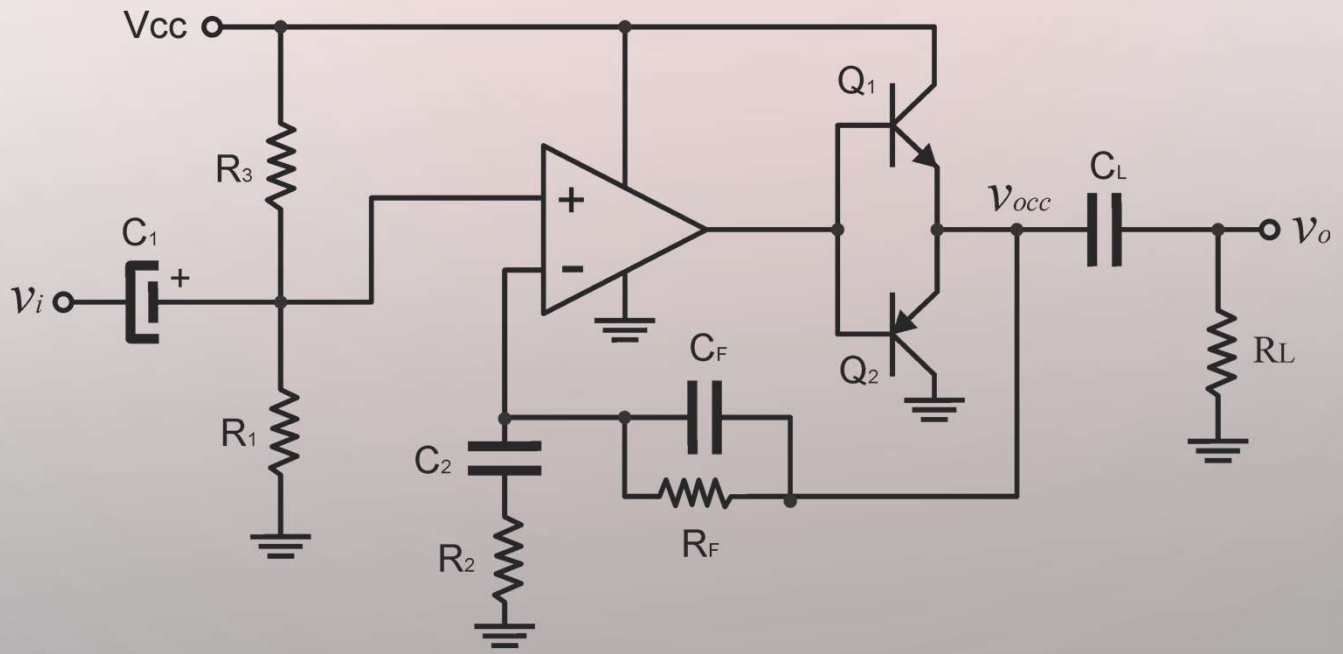
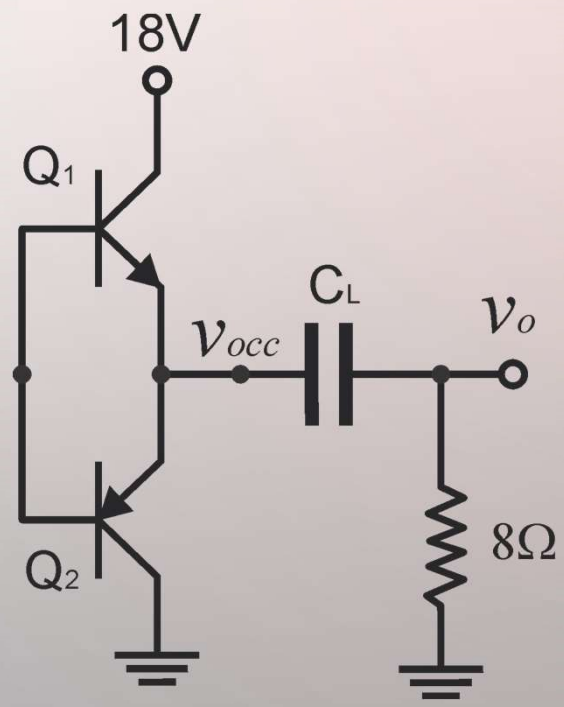
LABORATORIO N°2

AMPLIFICADORES CLASE-B

AMPLIFICADOR REALIMENTADO



DISEÑO DE LA SALIDA



Limite de excursión ideal:

$$\hat{v}_o = \hat{v}_o{}^{+} = \hat{v}_o{}^{-} = \frac{Vcc}{2} = 9V$$

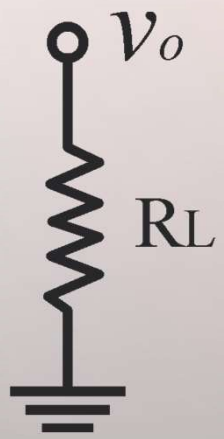
Potencia máxima en la carga:

$$P_L = \frac{\widehat{v_o}^2}{2R_L} = 5W$$

Potencia **máxima** en cada transistor:

$$P_{Q_1} = P_{Q_2} = \frac{\left(\frac{Vcc}{2}\right)^2}{\pi^2 R_L} = 2W$$

DISEÑO DE LA SALIDA – DISEÑO DE LA CARGA



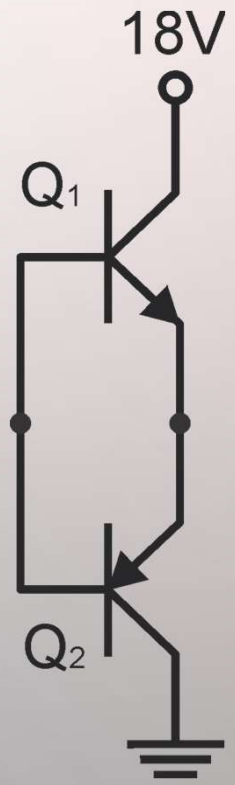
Diseño:

$$R_L = 8\Omega \text{ @ } 5\text{W}$$

Adoptamos:

$$R_L = 3x(2{,}7\Omega \text{ @ } 2\text{W}) = 8{,}1\Omega \text{ @ } 6\text{W}$$

DISEÑO DE LA SALIDA – DISEÑO DE Q_1 Y Q_2



Diseño:

$$V_{CE_{MAX}} > Vcc = 18V$$

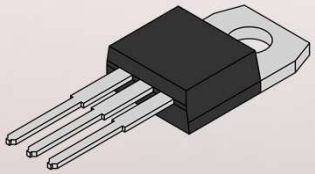
$$I_{C_{MAX}} > \frac{Vcc}{R_L} = 1{,}125A$$

$$P_D > 4 * P_{Q_{MAX}} = 8W$$

$$h_{FE_{MIN}} > I_{C_{MAX}}/10mA = 112{,}5$$

DISEÑO DE LA SALIDA – DISEÑO DE Q_1 Y Q_2

Adoptamos: TIP120 (NPN) / TIP125(PNP) – Darlington TO220



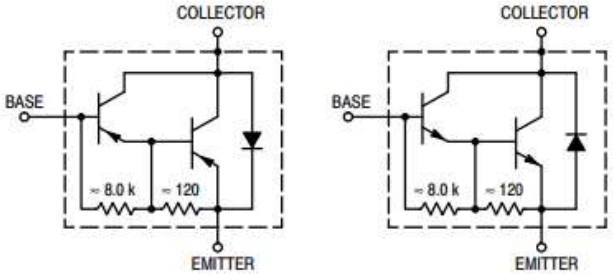
MAXIMUM RATINGS

Rating	Symbol	TIP120, TIP125	TIP121, TIP126	TIP122, TIP127	Unit
Collector–Emitter Voltage	V_{CEO}	60	80	100	Vdc
Collector–Base Voltage	V_{CB}	60	80	100	Vdc
Emitter–Base Voltage	V_{EB}	5.0			Vdc
Collector Current – Continuous – Peak	I_C	5.0 8.0			Adc
Base Current	I_B	120			mAdc
Total Power Dissipation @ $T_C = 25°C$ Derate above 25°C	P_D	65 0.52			W W/°C
Total Power Dissipation @ $T_A = 25°C$ Derate above 25°C	P_D	2.0 0.016			W W/°C
Unclamped Inductive Load Energy (Note 1)	E	50			mJ
Junction, Temperature Range	T_J, T_{stg}	−65 to +150			°C

THERMAL CHARACTERISTICS

Characteristic	Symbol	Max	Unit
Thermal Resistance, Junction–to–Case	$R_{\theta JC}$	1.92	°C/W
Thermal Resistance, Junction–to–Ambient	$R_{\theta JA}$	62.5	°C/W

TIP120, TIP121, TIP122 (NPN); TIP125, TIP126, TIP127 (PNP)

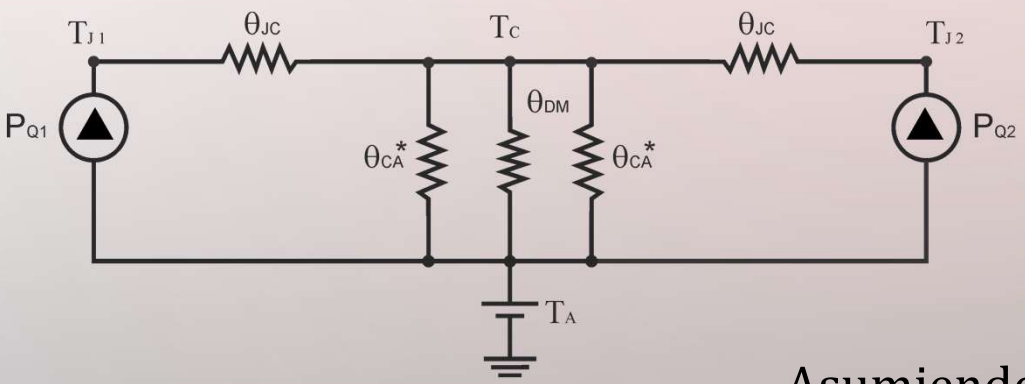


ON CHARACTERISTICS (Note 2)

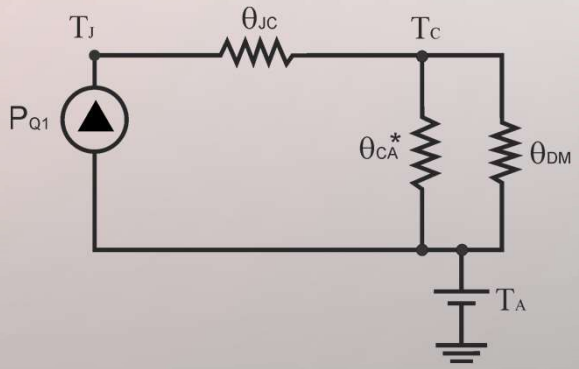
DC Current Gain (I_C = 0.5 Adc, V_{CE} = 3.0 Vdc) (I_C = 3.0 Adc, V_{CE} = 3.0 Vdc)	h_{FE}	1000 1000	– –	–
Collector–Emitter Saturation Voltage (I_C = 3.0 Adc, I_B = 12 mAdc) (I_C = 5.0 Adc, I_B = 20 mAdc)	$V_{CE(sat)}$	– –	2.0 4.0	Vdc
Base–Emitter On Voltage (I_C = 3.0 Adc, V_{CE} = 3.0 Vdc)	$V_{BE(on)}$	–	2.5	Vdc

DISEÑO DE LA SALIDA – DISEÑO TÉRMICO

Un disipador para ambos transistores



Dos disipadores



Asumiendo:

$$\theta_D // \theta_{CA}^* \approx \theta_D$$

$$\theta_D < \frac{T_J - T_A}{2P_Q} - \frac{\theta_{JC}}{2}$$

$$\theta_D < \frac{T_J - T_A}{P_Q} - \theta_{JC}$$

DISEÑO DE LA SALIDA – DISEÑO TÉRMICO

Diseño:

$$P_Q = 2W$$

Datasheet:

$$\theta_{Jc} = 1{,}92°/\mathrm{W}$$

Adoptamos:

$$T_J = 100°$$

$$T_A = 50°$$

Un disipador para ambos transistores:

$$\theta_{DM} < 11{,}5°/\mathrm{W}$$

Dos disipadores (uno para cada transistor):

$$\theta_{DM} < 23°/\mathrm{W}$$

<u>Se debe tener en cuenta la resistencia térmica del montaje</u>

$$\theta_{DM} = \theta_D + \theta_M$$

DISEÑO DE LA SALIDA – DISEÑO TÉRMICO

Encapsulado	Contacto Directo	Contacto directo y silicona	Contacto con mica sin silicona	Contacto con mica y silicona
TO5	1	0,7		
TO39	1	0,7	2	1,5
TO126	1,4	1	1,4	1,3
TO220	0,8	0,5	1,4	1,2
TO247AC	0,8	0,5	1,4	1,2
TO152	0,8	0,5	1,2	0,9
…	…	…	…	…

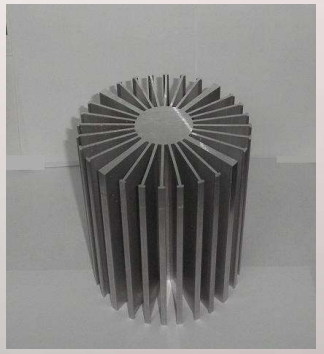
Un disipador para cada transistor:

$$\theta_{DM} < 23°/\mathrm{W}$$

$$\theta_D < 21{,}8°/\mathrm{W}$$

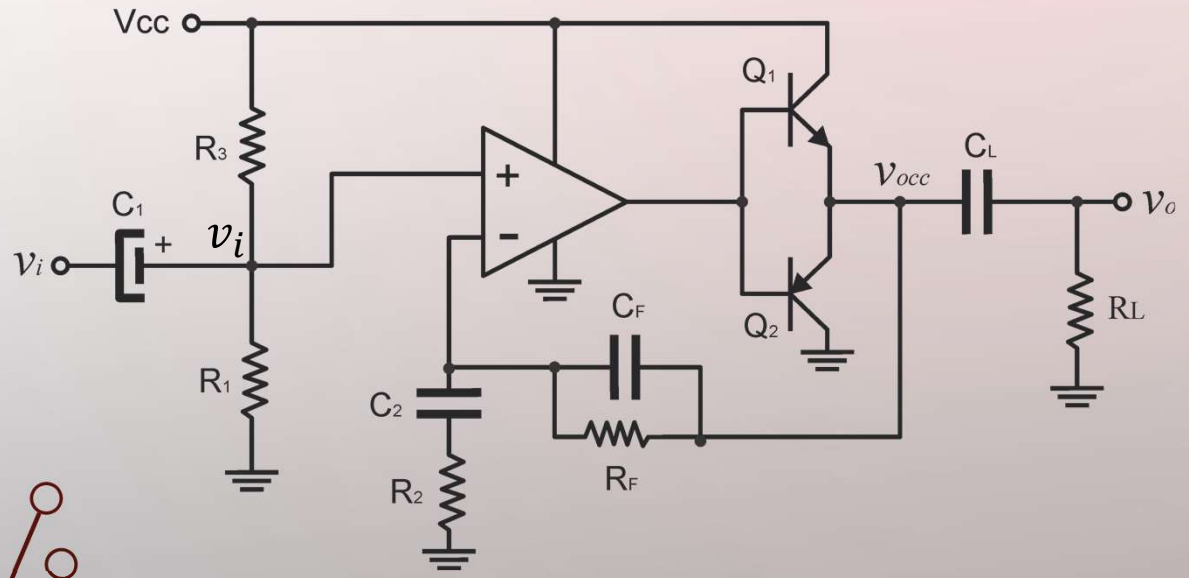
DISEÑO DE LA SALIDA – DISEÑO TÉRMICO

INTERNATIONAL ALU-EL (disipadores.com) $\theta_D < 21{,}8°/\mathrm{W}$



5235FD
20x20x20 1.5mm
21°/W

DISEÑO DE LA ENTRADA



$$\frac{v_o}{v_i}\Big|_{CC} = 1$$

$$z_i = R_1//R_2 = 20k\Omega$$

$$V_i = \frac{V_{CC}}{2}$$

Adoptamos:

$$R_1 = R_3 = 39k\Omega$$

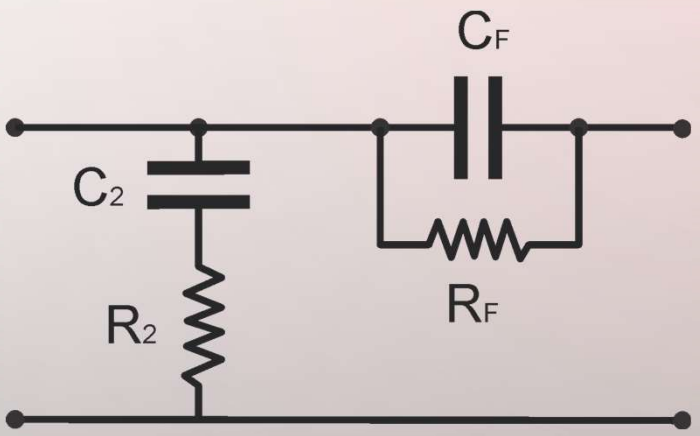
$$C_1 \gg \frac{1}{\omega_{\min} R_1//R_2} = 200nF$$

Adoptamos: $C_1 = 2{,}2\mu F$

R_1: fija de $39k\Omega$

R_3: $33k\Omega$ y un *preset* de $10k\Omega$

DISEÑO DE LA RED DE REALIMENTACIÓN



$$\left.\frac{v_o}{v_i}\right|_{f_{med}} = \frac{R_2 + R_F}{R_2} = 19 \Rightarrow R_F = 18 \cdot R_2$$

Adoptamos (se podría dejar un punto de ajuste):

$R_2 = 1k\Omega \qquad R_F = 18k\Omega$

$C_2 \gg \frac{1}{\omega_{\min} R_2}$ ➔ Adoptamos: $C_2 = 22\mu F$

$C_F = \frac{1}{2\pi\, 20kHz\, R_F} = 442pF$ ➔ Adoptamos: $C_F = 470pF$